List of Programs

1. Implement a program to define and use basic graphics primitives such as points, lines, and polygons.

2. Implement a program to draw a line using the Digital Differential Analyzer (DDA) algorithm.

3. Create a program to draw a line using Bresenham's line drawing algorithm.

4. Write a program to draw a circle using the Midpoint Circle algorithm.

5. Implement a program to draw a circle using Bresenham's circle drawing algorithm.

6. Create a program to perform 2D transformations: translation, rotation, and scaling on a given set of points.

7. Write a program to perform mirror reflection of a 2D shape across the x-axis and y-axis.

8. Create a program to draw basic two-dimensional objects like rectangles, triangles, and polygons using inbuilt functions..

9. Write a program to implement the Cohen-Sutherland line clipping algorithm to clip a line within a rectangular clipping window.

10. Write a program to apply a shearing transformation to a 2D object (like a square or triangle) and display the original and transformed objects.

11. Write a program to apply various coloring techniques to 2D pictures.

12. Write a program to implement line clipping using the Cohen-Sutherland line clipping algorithm.

13. Write a program to perform 3D transformations: translation, rotation, and scaling on a set of 3D points.

14. Implement a program to demonstrate simple animations using transformations such as translation, rotation, and scaling.

15. Write a program to apply various coloring techniques to 3D object.

16. Write a program to draw a simple hut with using lines, shapes and coloring techniques to 3D object.

17. **Implement a 3D Projection program** to convert 3D coordinates into 2D screen coordinates using both Parallel (Orthographic) and Perspective projections.

18. **Create a program to implement a simple Illumination Model**, such as the Ambient, Diffuse, and Specular (Phong) lighting model on a 3D sphere.

19. **Write a program to draw a Bezier Curve** using four control points to understand parametric curves.

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